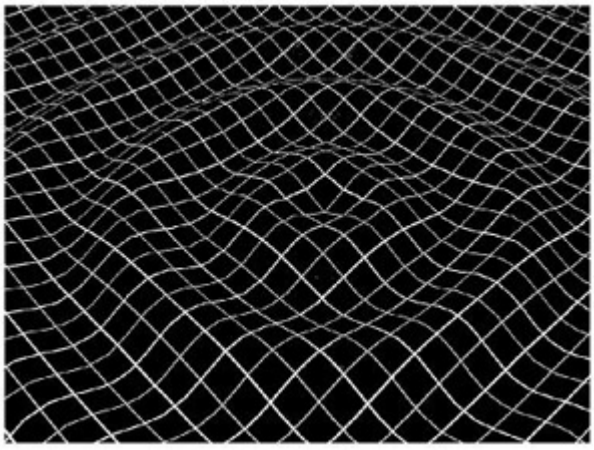


Figure 14.2 Gravitational waves are ripples in the fabric of spacetime.

With every passing peak and trough, a gravitational wave's distorted geometry would stretch space—and everything in it—in one direction, and then compress space—and everything in it—in a perpendicular direction, as in the highly exaggerated depiction in Figure 14.3. In principle, you could detect a gravitational wave's passing by repeatedly measuring distances between a variety of locations and finding that the ratios between these distances had momentarily changed.

In practice, no one has been able to do this, so no one has directly detected a gravitational wave. (However, there is compelling, indirect evidence for gravitational waves.[4]) The difficulty is that the distorting influence of a passing gravitational wave is typically minute. The atomic bomb tested at Trinity on July 16, 1945, packed a punch equivalent to 20,000 tons of TNT and was so bright that witnesses miles away had to wear eye protection to avoid serious damage from the electromagnetic waves it generated. Yet, even if you were standing right under the hundred-foot steel tower on which the bomb was hoisted, the gravitational waves its explosion produced would have stretched your body one way or another only by a minuscule fraction of an atomic diameter. That's how comparatively feeble gravitational disturbances are, and it gives an inkling of the technological challenges involved in detecting them. (Since a gravitational wave can also be thought of as a huge number of gravitons traveling in a coordinated manner—just as an electromagnetic wave is composed of a huge number of coordinated photons—this also gives an inkling of how difficult it is to detect a *single* graviton.)